Industrial A5S Sensor with Modbus TCP/MQTT Communication

1. Product Overview

WIN-SN-A5S-MTCP-WiFi is a versatile data acquisition device designed for real-time monitoring in industrial applications, especially ducts. It supports a wide range of external sensors with **UART interface**, providing flexibility for various measurements. Offering reliable communication and long-range data transfer over several meters, making it ideal for large-scale installations.

Its robust design makes it suitable for **installations in HVAC systems, environmental monitoring, and process control**. With **configurable data sampling rates, multi-sensor compatibility, and easy configuration**, WIN-SN-A5S-MTCP-WiFi ensures **accurate Sensor (CH4/H2S/CO2/CO/NH3/PH3/C2H4) data acquisition and reliable performance** in demanding industrial environments, making it an excellent choice for **industrial and IOT applications**. It supports multiple communication protocols, allowing users to select either TCP (**Modbus TCP**) or Wi-Fi (**MQTT**) based on their application requirements, providing greater flexibility for industrial and IOT deployments. Additionally, it features a 4-digit 7-segment display that provides real-time measurement data, allowing users to monitor sensor readings directly without requiring an external interface.

Sensor Parameters

	SN	Sensor Name	Measuring Parameter	Measuring Range	Output	Operating Temperature
☐	**a**	**A-5S-CO**	CO	0-1000 ppm	TCP/WiFi	-20°C to +50°C
☐	**b**	**A-5S-CO2**	CO2	0-5000/ 10000/ 40000 ppm	TCP/WiFi	-20°C to +50°C
☐	**c**	**A-5S-NH3**	NH3	0-100 ppm	TCP/WiFi	-20°C to +50°C
☐	**d**	**A-5S-PH3**	PH3	0-1000 ppm	TCP/WiFi	-20°C to +50°C
☐	**e**	**A-5S-CH4**	CH4	0-10000/ 20000 ppm	TCP/WiFi	-20°C to +50°C
☐	**f**	**A-5S-H2S**	H2S	0-100 ppm	TCP/WiFi	-20°C to +50°C
☐	**g**	**A-5S-C2H4**	C2H4	0-20 ppm	TCP/WiFi	-20°C to +50°C

	Product Configuration
☐	A5S+MTCP
☐	A5S+MTCP+WiFi
☐	A5S+MTCP+WiFi+D

2. Technical Parameters

a) CO Sensor

- Target Gas: Carbon Monoxide(CO)
- Detection Range: 0 to 1000ppm
- Accuracy: ±1 ppm or ±1%
- Detection Accuracy: ~90%
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

b) CO2 Sensor

- Target Gas: Carbon Dioxide (CO2)
- Detection Range: 0 to 5000/10000/40000 ppm
- Accuracy: ±1 ppm or ±1%
- Detection Accuracy: ~90%

- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

c) NH3 Sensor

- Target Gas: Ammonia (NH_3)
- Detection Range: 0 to 100ppm
- Accuracy: ±1 ppm or ±1%
- Detection Accuracy: ~90%
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

d) PH3 Sensor

- Target Gas: Phosphine (PH_3)
- Detection Range: 0 to 1000ppm
- Accuracy: ±1 ppm or ±1%
- Detection Accuracy: ~90%
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

e) CH4 Sensor

- Target Gas: Methane (CH4)
- Detection Range: 0 to 10000/20000 ppm
- Accuracy: ±50 ppm or ±1%
- Detection Accuracy: ~90%
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

f) H2S Sensor

- Target Gas: Hydrogen Sulfide (H2S)
- Detection Range: 0 to 100 ppm
- Accuracy: ±1 ppm or ±1%
- Detection Accuracy: ~90%
- Technology: Non-Dispersive Infrared (NDIR)
- Response Time: Less than 3 seconds

g) C2H4 Sensor

- Target Gas: Ethylene (C2H4)
- Detection Range: 0 to 20 ppm

- Accuracy: Resolution of 0.5 ppm (Zero drift 0~1 ppm)
- Detection Accuracy: Repeatability < ±2% output signal
- Technology: 3 Electrodes Electrochemical
- Response Time: Less than 90 seconds

Note: All sensors require a 90-second warm-up period after power-on to provide accurate measurements.

2. Sensor Specifications

- **Wide temperature range:** −40 °C to +125 °C
- **Full humidity range:** 0 % to 100 % RH
- **Power supply DC:** 12-24V@1A.
- **Product dimension:** 85*85*36 mm.
- **Certifications:**

3. Power Supply Connection

DC Connection

- Use 12-24V@ 1A, Power supply.
- Don't make loose connection.

5. Recommended Cable Electrical Characteristics

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

6. Device Configuration Process

1. Connect the Device

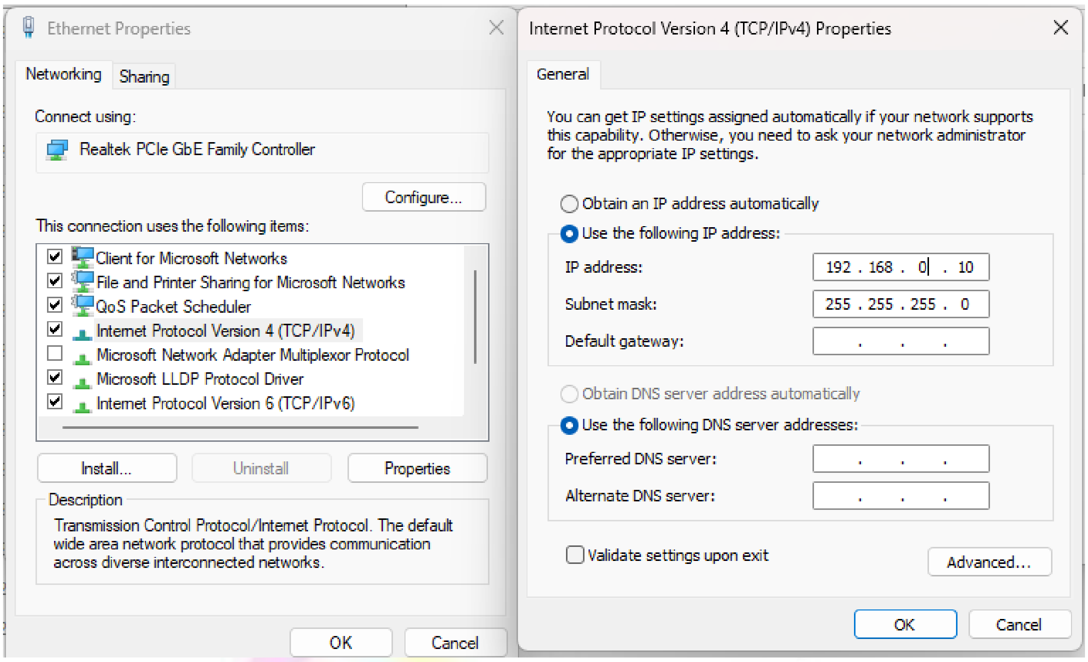
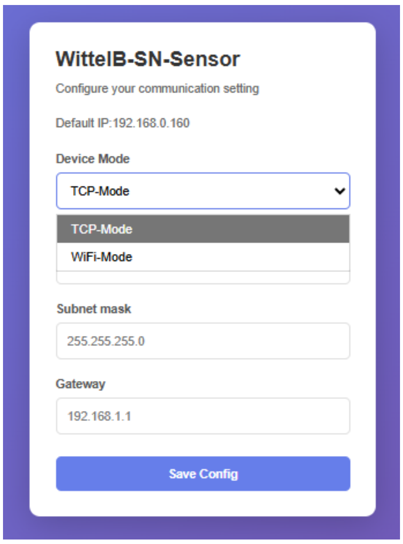
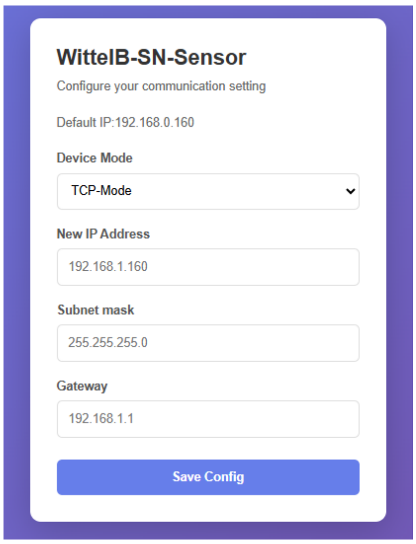


Image: 1

- Connect 12-24 VDC-1A DC adaptor to power the board
- Make the IPV4 configuration as 192.168.0.xx
- Enter 192.168.0.160 in web interface

1. Mode Selection

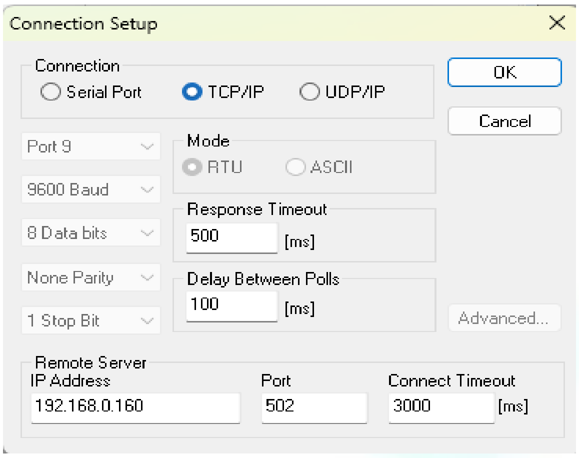


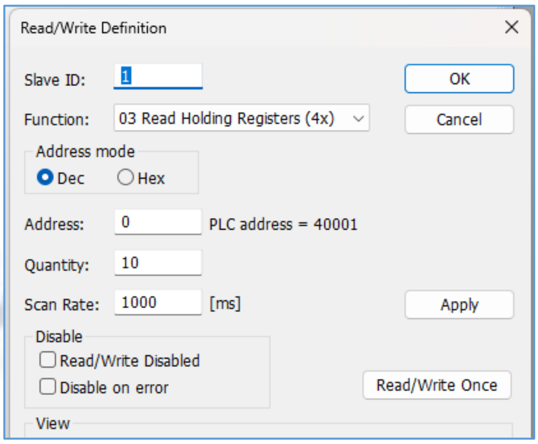


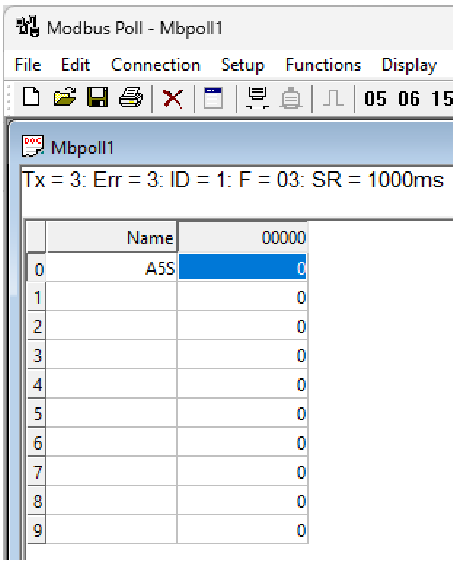
- Enter **192.168.0.160** in the browser search bar and select the **Device Mode**.
- Select the device mode as **TCP-Mode** to get the sensor data via **Modbus TCP**, or select **WiFi Mode** to get the sensor data in the **MQTT cloud** through the local WiFi network.

2. TCP Mode

- Here is an example of receiving data on Mb poll software.
- Open Mb-poll software connect using configuration setting (Default IP 192.168.0.160).
- After connecting you will get data as shown below.







- Above image shows sensor data over Modbus TCP with default Modbus settings.
- Register 40001 contains the measured gas concentration value corresponding to the connected sensor (CO/CO_2/NH_3/PH_3/H_2S/CH_4/C2H4).
- Keep the jumper in the **single/open** as shown in image 1.2 position to change the IP address.
- Enter the desired IP address on the web page and click **Save**. The new IP address will be applied automatically.
- Connect using the new IP address to access the device and receive the data.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	A5S Values	40001	0x03	TCP/IP	16-bit int

Modbus Registers Mapping

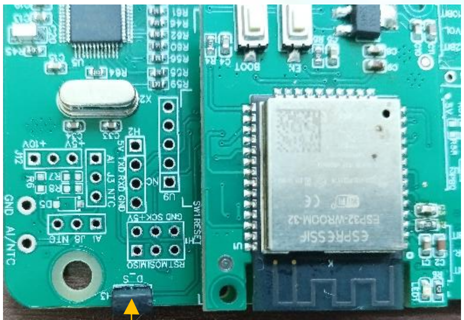


Image: 1.1

Image: 1.2

Hard reset Setting

- Somehow you forgot the baud rate or slave id of the board then Place the jumper (as shown in the above image 1.1) from the board reboot the system or press the reset button (given on the board). Then you will get the default baud rate of 9600 and slave id 1.

3. WIFI Mode

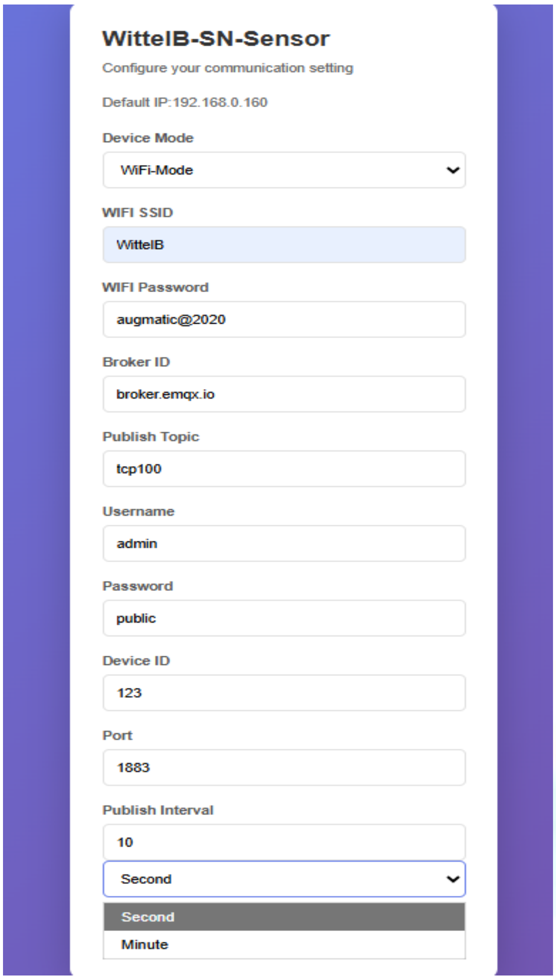


Image: 2 **Image: 3**

- Enter the local WiFi credentials as shown in above image2 so the device can connect to that WiFi network. Using this connection, it will send the sensor data to the cloud.
- Then enter the required MQTT Credentials and make sure the credentials should be correct.
- As shown in Image 2, you can select the publish interval in either minutes or seconds. MQTT data will be published according to the selected interval.
- Enter the MQTT credentials to transmit the sensor data to the cloud.
- Image 3 shows the sensor data at cloud for the selected topic.
- The published JSON payload includes a real-time timestamp, allowing each sensor reading to be tracked with its exact time.

Contact Information

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